

NEW YORK UNIVERSITY

04/09 decipEHR Project

Jacob (Jiajian) Ma

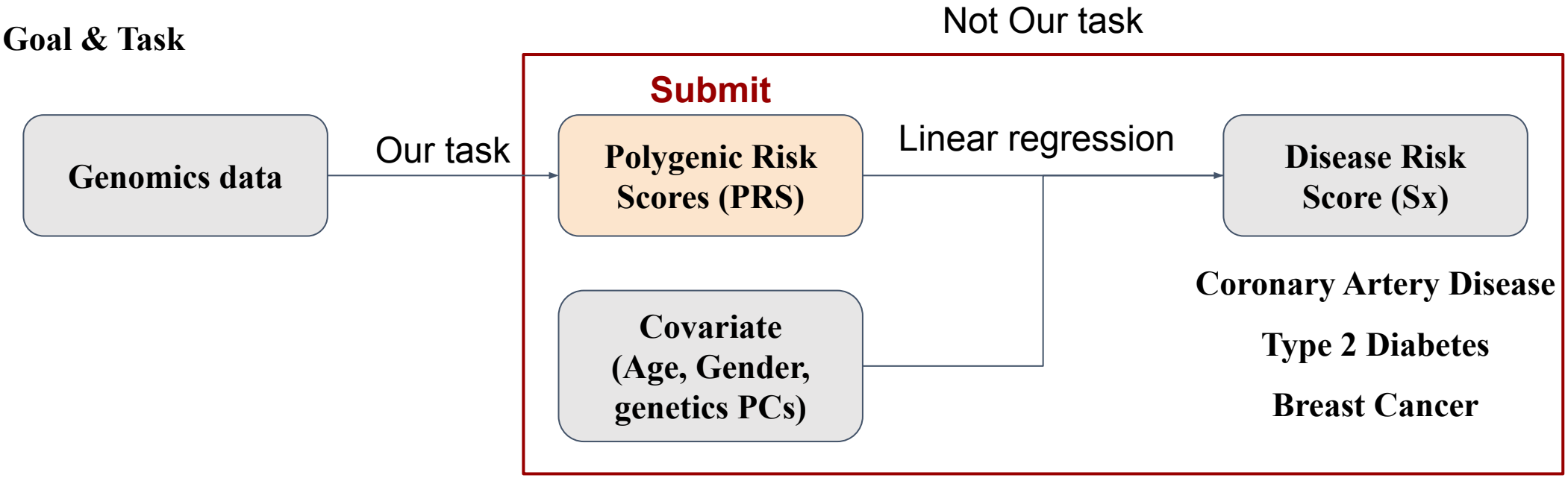
2026/04/09

deciphEHR Project (BioDash)

- Introduction
- Data
- Proposal
- Plan

Introduction

Goal & Task

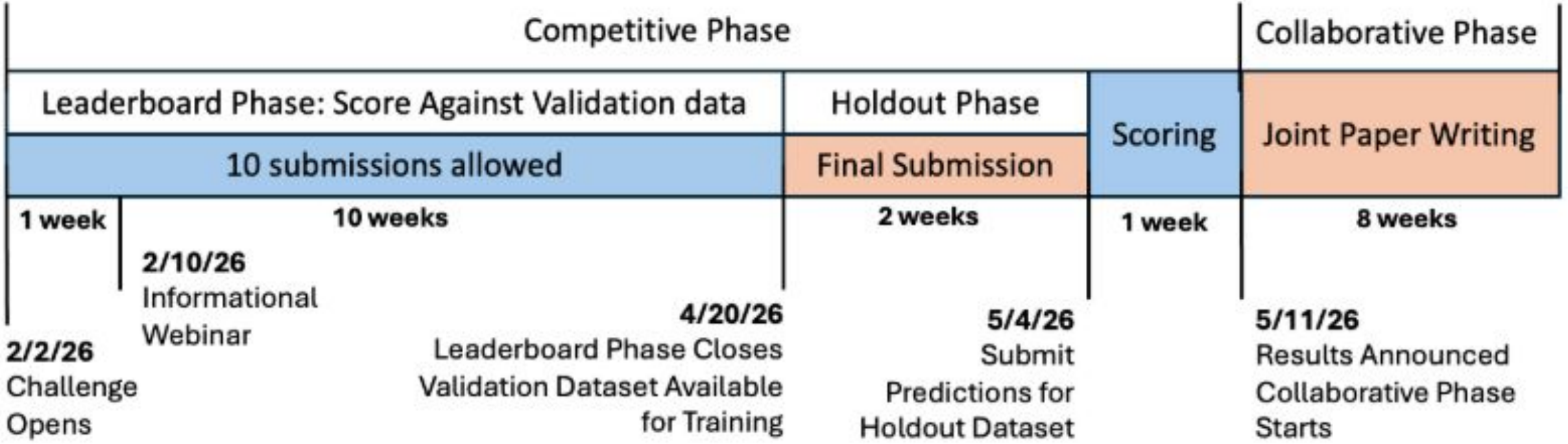


Linear Model Built by Organizer

- *Null Model (X=0),* prevalence of the phenotype in training set
- *Covariate Only Model (X=C), Model only use covariate* $\beta_0^C + \beta_{\text{age}}^C \cdot \text{Age} + \beta_{\text{sex}}^C \cdot \text{Sex} + \sum_{j=1}^{10} \beta_{\text{PC}_j}^C \cdot \text{PC}_j \equiv S_C$
- *PRS-Only Model (X=G), Model only use PRS* $\beta_0^G + \beta_{\text{PRS}}^G \cdot \text{PRS} \equiv S_G$
- ***Full Model (X=C, G),*** *use both covariate and PRS* $\beta_0^F + \beta_{\text{PRS}}^F \cdot \text{PRS} + \beta_{\text{age}}^F \cdot \text{Age} + \beta_{\text{sex}}^F \cdot \text{Sex} + \sum_{j=1}^{10} \beta_{\text{PC}_j}^F \cdot \text{PC}_j \equiv S_F$

Introduction

- <6 weeks before Final Submission



Data

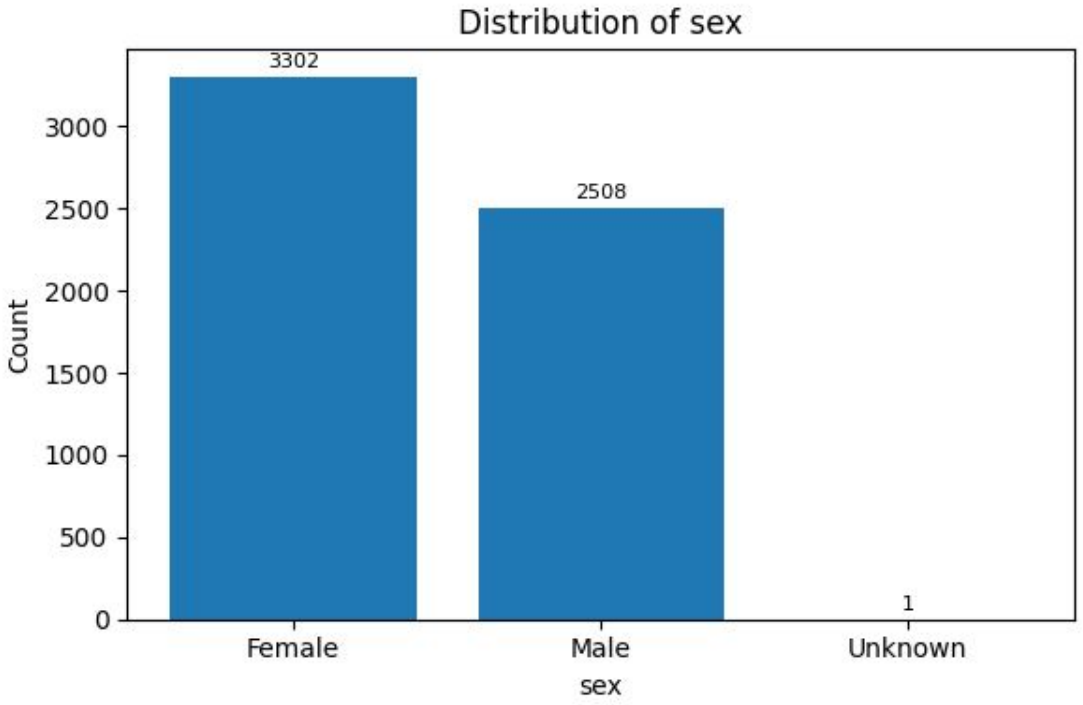
- Organization
  - Genotype Files
  - Supplementary Files
  - Phenotype Files

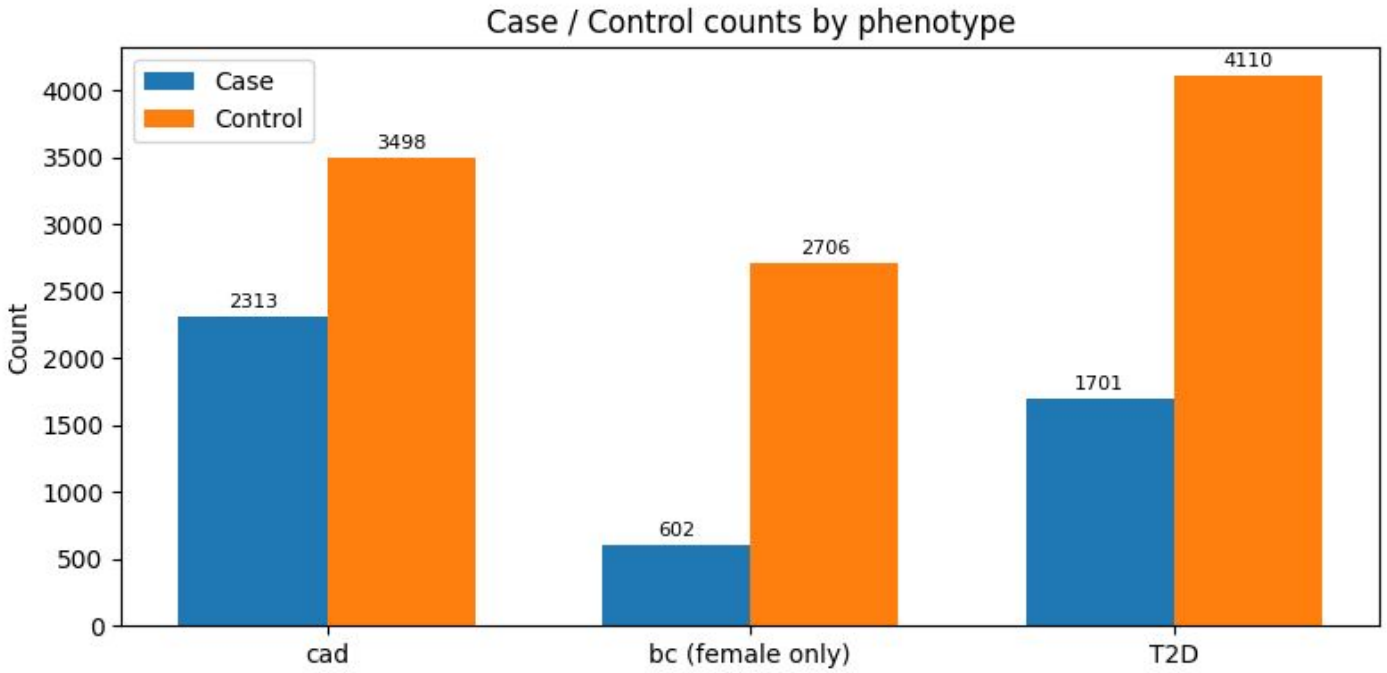
```
/gpfs/data/deciphEHRlab/deciphEHRdream/training_phase/
├── genotype_files/
│   ├── vcf_files/
│   │   ├── chr*.vcf.gz
│   │   └── chr*.vcf.gz.tbi
│   └── plink_files/
│       ├── chr*.pgen
│       ├── chr*.pvar
│       └── chr*.psam
├── supplementary_files/
│   ├── dbsnfp/
│   ├── info_scores/
│   ├── snpeff/
│   └── summary_statistics/
└── phenotype_files/
    ├── Training_Cov_file.txt
    └── Testing_Cov_file.txt
```

Data (Phenotype & covariates)

- Size: Training (5,811) + Validation (2,852) + Testing (~3000)

	sampleid	sex	age_sd	PC1_sd	PC2_sd	PC3_sd	PC4_sd	PC5_sd	PC6_sd	PC7_sd	PC8_sd	PC9_sd	PC10_sd	cad	bc	T2D
0	S-781185	Male	1.288303	-0.555962	0.480146	0.224748	0.241001	0.320088	1.938814	0.348092	0.387093	0.610129	-0.450001	1	0	0
1	S-312853	Female	1.591000	-0.482392	0.222560	-0.349030	-0.697747	-0.539562	0.463947	-0.214933	0.386187	-0.588972	-0.390756	0	0	0
2	S-879678	Male	0.077515	0.154176	-3.259585	1.898367	-0.280359	-0.868405	0.188799	2.969257	-4.082796	1.091930	1.139250	1	0	1
3	S-964332	Female	-0.770037	1.364458	0.080234	0.323463	-0.224608	1.842692	-0.409662	-1.222077	0.806165	-0.700859	0.216519	0	0	0
4	S-734060	Male	0.440751	-0.562651	0.332478	-0.273394	-1.414405	-0.311043	-0.842868	-0.864892	-0.473900	-1.954522	1.386096	1	0	0





Data (Genotype)

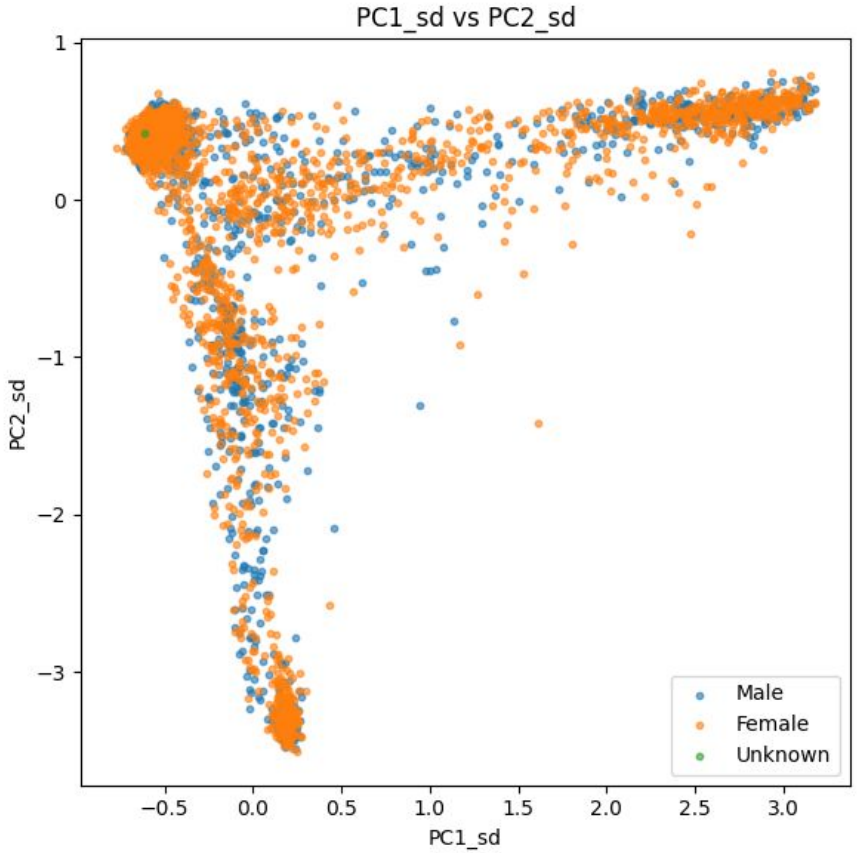
- # Variants
  - ~111M in total
- Genotype Files
  - vcf
  - plink
    - .pgen (main data)
    - .pvar (meta data for variants)
    - .psam (meta data for patient)

Data

- **Supplementary Files (Useful in QC & annotation [feature selection / Enhancement])**
  - **Info Scores**
  - **Summary Statistics**
  - **snpeff**
  - **dbsnfp**

```
/gpfs/data/deciphEHRlab/deciphEHRdream/training_phase/
├── genotype_files/
│   ├── vcf_files/
│   │   ├── chr*.vcf.gz
│   │   └── chr*.vcf.gz.tbi
│   └── plink_files/
│       ├── chr*.pgen
│       ├── chr*.pvar
│       └── chr*.psam
├── supplementary_files/
│   ├── dbsnfp/
│   ├── info_scores/
│   ├── snpeff/
│   └── summary_statistics/
└── phenotype_files/
    ├── Training_Cov_file.txt
    └── Testing_Cov_file.txt
```

Overall Pipeline

Genomics data → **Quality Control (Filter unreliable)** → 

→ **ıkage** Count

non-annotated method

- **LD information**
- **LD + Functional information**

annotated method

Question:

1. Standard Repo for QC ?
   a. filter for info score (0-1, 0.3-0.8)
   b. frequency (depends on disease)
2. Need to model multi-ancestry ? Hows ? Especially for using external data

	#CHROM	ID	REF	ALT	AI
0	1	c1_10454_snv_C_T	C	T	
1	1	c1_14521_snv_C_T	C	T	
2	1	c1_14610_snv_T_C	T	C	

SOTA methods

nature communications

Article

https://doi.org/10.1038/s41467-026-68599-7

Comprehensive benchmarking single and multi ancestry polygenic score methods with the PGS-hub platform

Received: 4 March 2025

Accepted: 12 January 2026

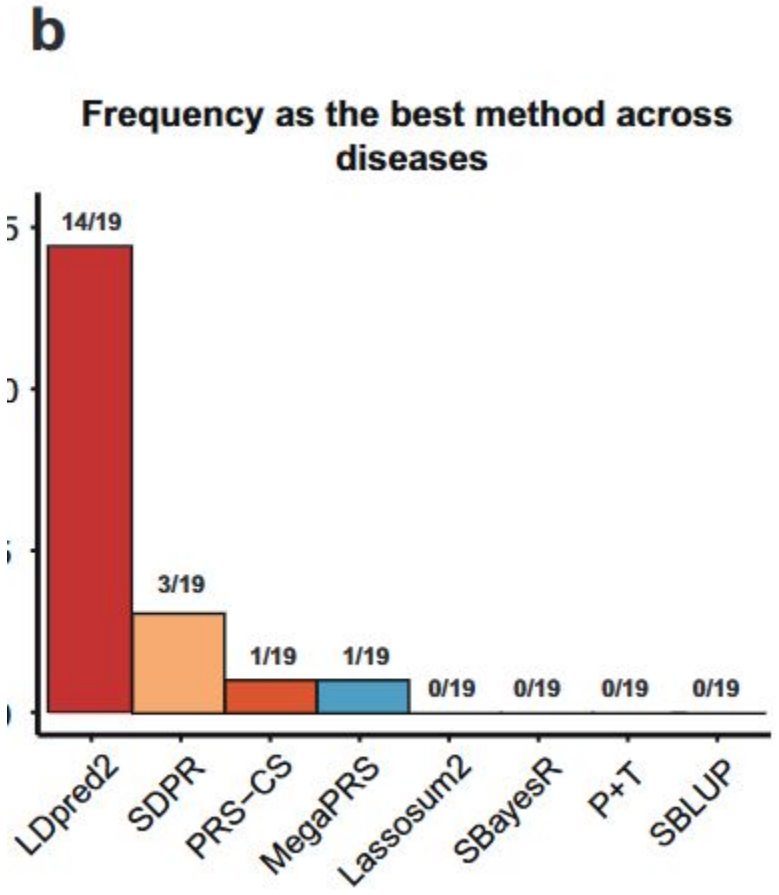
Published online: 25 January 2026

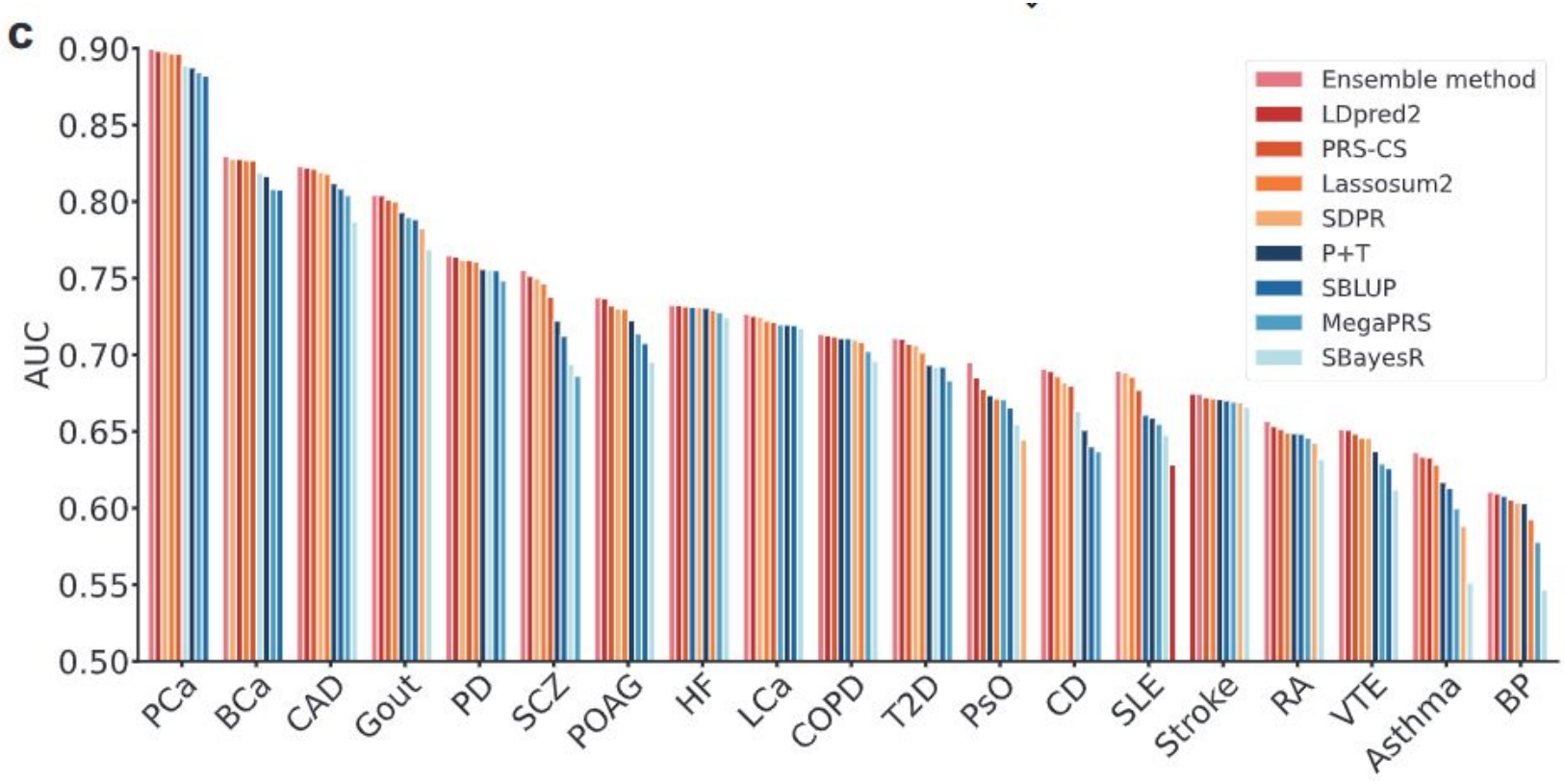
Check for updates

Xingyu Chen[1,2,3,4,5,9], Fei Wang[1,2,9,10] ✉, Hongqiang Zhao[1,2,9], Jing Hao[1,2,3], Yunga A[1,2,3], Xiong Yang[1,2], Tingfeng Xu[1,2,3], Yubo Zhou[1,2], Qiuli Chen[1,2,3], Rufan Zhang[1,2], Kang Yu[1,2,3], Komal Zaib[1,2,3], Akl C. Fahed[4,5,6,7], Guangyao Zhai[8] ✉ & Minxian Wang[1,2,3,10] ✉

SOTA methods

SOTA non-annotation method: LDpred2 & Ensemble method always better (LDpred2, 2021)





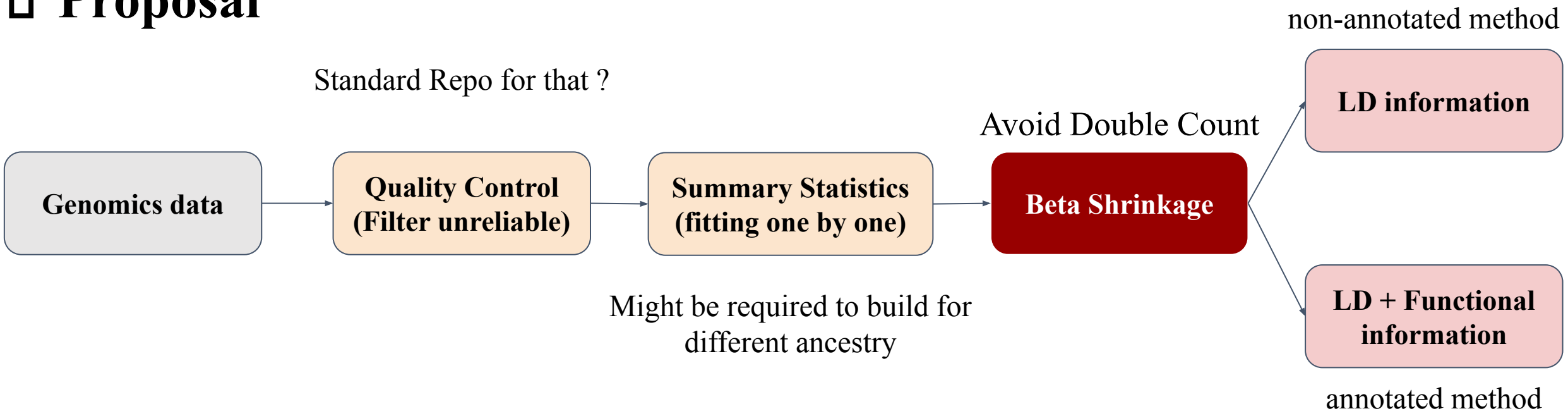
SOTA methods

SOTA annotation method: **MegaPRS (2021)** and **SBayesRC (2024)**

- Better than non-annotation methods
- Functional info make sense

MegaPRS non-annotation counterparts, achieving superior performance relative to the best-performing non-annotation methods (LDpred2, PRS-CS, Lassosum2, and SDPR) for 12 out of 19 disease traits and 8 out of 17 quantitative traits (Supplementary Tables 29 and 30). Similarly, another PGS method—leveraging functional genomic annotations—SBayesRC, according to our supplementary analysis, showed superior performance to many existing non-annotation methods (five diseases and seven quantitative traits) (Supplementary Tables 31 and 32). However, since this version of PGS-hub focuses on foundational PRS models without functional annotations, functional-enabled PGS methods will be systematically integrated in subsequent versions of PGS-hub.

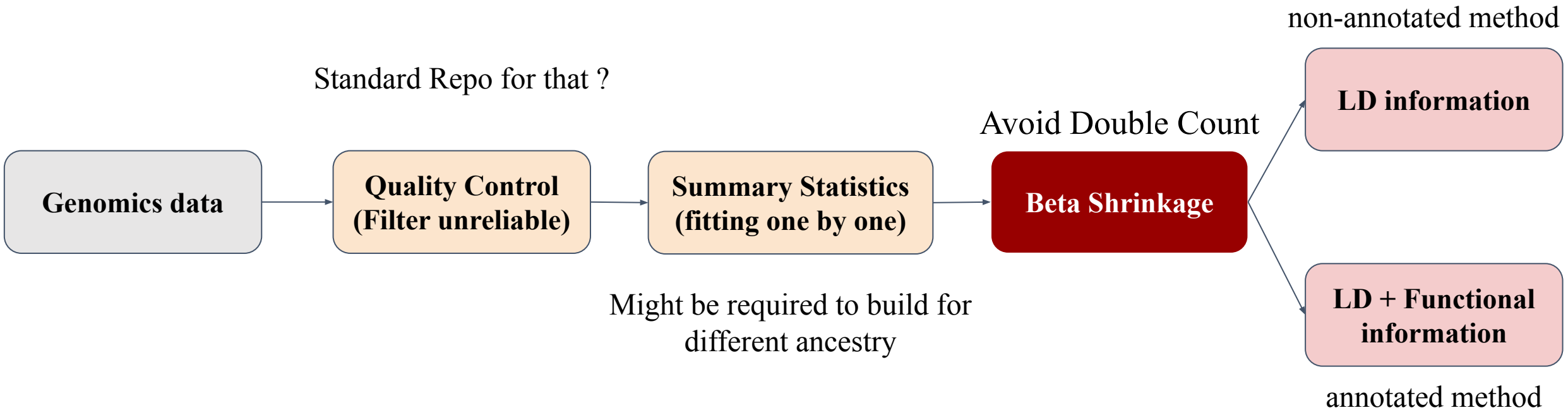
Proposal



For this competition: Ensemble of annotated & non-annotated method

- **Baseline**
  - Ensemble of non-annotated method
  - Ensemble of annotated & non-annotated method
- **Potential improvement (coding language is a problem)**
  - Better functional annotation in non-coding variants (AlphaGenome)
  - Personalize functional annotation (AlphaGenome)

Proposal



No representation learning method built in this field yet !

- AlphaGenome: gene sequence → representation for biochemical behaviour
- Objective: variant and ref gene context → representation for disease risk / physiological behaviour (possible ?)
- Common Pre-Training methods doesn't work
- （genetic data as context and pretrained with next token prediction of ICD-10 codes? ）

Competition Metrics

- **Metrics (*Only Full Model* used in ranking in this Competition)**
  - **Odd ratio (Main metrics): Ranking Ability**
    - **Odd: P(y=1 | Condition) / P(y=0 | Condition) → Relative Prevalence under certain condition**
    - **Odd ratio: $\text{Odd}_{\text{condition_1}}$ / $\text{Odd}_{\text{condition_2}}$**
    - **Two Conditions for this task:**
      - **Sx > 90%**
      - **Sx ≤ 90%**

$$\text{OR}_X = \frac{Odds(S_X > q^X_{90\%})}{Odds(S_X \le q^X_{90\%})} \qquad Odds(S_X > q^X_{90\%}) = \frac{\text{Prob}(y = 1 | S_X > q^X_{90\%})}{\text{Prob}(y = 0 | S_X > q^X_{90\%})} \qquad Odds(S_X \le q^X_{90\%}) = \frac{\text{Prob}(y = 1 | S_X \le q^X_{90\%})}{\text{Prob}(y = 0 | S_X \le q^X_{90\%})}$$

  - **The Nagelkere R2 metric (Secondary): Relative Fitting Ability compared with Null Model**
    - **L0 is the likelihood of null model; Lx is the likelihood of the target model**

$$R^2_{\text{Nag}} = \frac{1 - \left(\frac{L_o}{L_x}\right)^{\frac{2}{N}}}{1 - (L_0)^{\frac{2}{N}}}$$

Final Performance was obtained by Running Bootstrap (1000 times) in Test data

Current Leaderboard

		Full Model (PRS and Covariates)		Genetics Model (PRS Only)	
Submission File	**Submission Date**	**OR**	R^2	**OR**	R^2
KyDeciph_cad_3.txt	2026/3/17	7.280	0.311	1.780	0.027
Covariates Only Model	-	5.830	0.297	-	-
Reference Model	-	5.590	0.297	0.930	0.001
Random_cad_1.txt	2026/3/18	5.590	0.297	0.750	0.000

		Full Model (PRS and Covariates)		Genetics Model (PRS Only)	
Submission File	**Submission Date**	**OR**	R^2	**OR**	R^2
KyDeciph_T2D_2.txt	2026/3/17	4.510	0.168	1.850	0.052
Reference Model	-	3.250	0.131	0.800	0.001
Random_T2D_1.txt	2026/3/18	3.200	0.130	1.010	0.000
Covariates Only Model	-	3.200	0.130	-	-

		Full Model (PRS and Covariates)		Genetics Model (PRS Only)	
Submission File	**Submission Date**	**OR**	R^2	**OR**	R^2
Reference Model	-	2.390	0.070	2.020	0.020
KyDeciph_bc_1.txt	2026/3/17	2.230	0.070	1.570	0.014
Random_bc_1.txt	2026/3/18	1.630	0.047	0.830	-0.001
Covariates Only Model	-	1.400	0.048	-	-

To do list

- Finish Quality Control
- Establish Baseline
  - non-annotated methods
  - annotated methods



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Thanks for Listening !

Jacob (Jiajian) Ma